1.

Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

ANS:-

Static IP Address Assignment:

Device	Static IP Address	Subnet Mask
Laptop 1	192.168.1.10	255.255.255.0
Laptop 2	192.168.1.11	255.255.255.0
Laptop 3	192.168.1.12	255.255.255.0
Laptop 4	192.168.1.13	255.255.255.0
Laptop 5	192.168.1.14	255.255.255.0

Connectivity Verification The ping command was used between the devices. All devices responded successfully, confirming that they are connected and working in the same subnet.

2.

Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

ANS:-

DHCP Server Configuration:

- DHCP Configuration: A DHCP server was configured with the IP range 192.168.1.10-192.168.1.50. Three devices were set to obtain their IP addresses automatically.

Device	Assigned IP Address
PC 1	192.168.1.10
PC 2	192.168.1.11
PC 3	192.168.1.12

- Result: All three devices received unique IP addresses automatically from the correct DHCP range, confirming that the DHCP server is working properly.

3.

Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface. Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

ANS:-

Local DNS Configuration:

- **Configuration:** A DNS entry was created to map musicplayer.local to 192.168.1.50.
- **Verification:** The hostname musicplayer.local was entered in the browser, and it successfully opened the local web page/device interface at 192.168.1.50.
- **Result:** The DNS configuration worked correctly, allowing the local device to be accessed using a custom hostname instead of its IP address.

4.

Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN. Hint: Use ping or file sharing to test connectivity between devices in different VLANs.

ANS:-

VLAN Configuration:

- VLAN Setup:

VLAN	Name	Devices
VLAN 10	Workstations	PC1, PC2
VLAN 20	Smart Devices	PC3, PC4

- Connectivity Test:
  - $PC1 \rightarrow PC2$: Ping successful
  - $PC3 \rightarrow PC4$: Ping successful
  - $PC1 \rightarrow PC3$: Ping failed
  - $PC2 \rightarrow PC4$: Ping failed
- **Result:** Devices in the same VLAN can communicate, while devices in different VLANs cannot communicate without inter-VLAN routing.

5.

Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

ANS:-

NAT Configuration:

- The router was configured with NAT to translate private LAN IP addresses into the router's public IP address.
- **Testing:** A LAN device was used to access an external server such as www.wikipedia.org. The connection was successful, confirming that NAT was working correctly.
- **Result:** NAT successfully allowed devices on the LAN to access the external network while using private IP addresses internally.